Machine Learning-Based Analysis of Gene Expression Profiles for Cancer and Liver Disease Classification



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This dissertation is submitted for the degree of
Bachelor of Science in Engineering (B.Sc. Engg.) in Computer Science and Engineering

Report Code:

BGC TRUST UNIVERSITY BANGLADESH

Department of Computer Science and Engineering

7-th Semester B.Sc. Engineering Examination july-dec 26

Course-No:CSE 070613-project701

Title: "Machine Learning-Based Analysis of Gene Expression Profiles for Cancer and Liver Disease Classification"

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Session: july-dec 26

Submission Date: 5 January 2026

APPROVAL FOR SUBMISSION

This is to certify that the thesis proposal entitled "Machine Learning-Based Analysis of Gene Expression Profiles for Cancer and Liver Disease Classification" submitted by Md. Tahsin Islam Chowdhury (ID: 230240012) to the Department of Computer Science and Engineering, BGC Trust University, has been reviewed and approved as a partial fulfillment of the requirements for the Bachelor of Science (B.Sc.) in Computer Science and Engineering. The proposed research has been prepared under my supervision and is considered suitable for submission and further evaluation

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Declaration

I hereby declare that this thesis proposal entitled “Machine Learning-Based Analysis of Gene Expression Profiles for Cancer and Liver Disease Classification” is my own original work and has been prepared under the supervision of Md. Asaduzzaman, Department of Computer Science and Engi neering, BGC TRUST UNIVERSITY BANGLADESH . I further declare that this proposal has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or other academic qualification. All sources of information, ideas, and materials used in this proposal have been properly acknowledged and cited in accordance with academic standards. I understand that any form of plagiarism or academic misconduct may result in disciplinary action in accordance with the rules and regulations of BGC Trust University of Bangladesh.

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Acknowledgements

I would like to express my sincere gratitude to my supervisor, Md. Asaduzzaman, Lecturer, Department of Computer Science and Engineering, BGC Trust University Bangladesh, for his invaluable guidance, continuous encouragement, and constructive suggestions throughout the completion of this research. His support and expertise have greatly contributed to the successful development of this thesis. I am also grateful to the Department of Computer Science and Engineering, BGC Trust University Bangladesh, for providing the academic environment and resources necessary for this study. Furthermore, I would like to thank my teachers, classmates, and friends for their encouragement and valuable support. Finally, I express my deepest gratitude to my parents and family members for their unwavering love, patience, and constant motivation, which inspired me to complete this research successfully

Abstract

Advancements in genomic data science and machine learning have created new opportunities for understanding complex diseases through the analysis of large-scale gene expression data. Cancer and liver disease are among the leading causes of mortality worldwide, and early identification of disease-associated molecular patterns is essential for improving diagnosis and treatment. This research proposes a comparative genomic data science approach to classify cancer and liver disease using publicly available gene expression datasets obtained from the National Center for Biotechnology Information (NCBI) Gene Expression Om nibus (GEO). The selected datasets include GSE329336 for cancer and GSE280687 for liver disease. The proposed methodology consists of data preprocessing, dataset splitting, machine learning model development, model training, and performance evaluation. During preprocessing, duplicate samples will be removed, missing values will be handled, the data will be normalized, and feature selection techniques will be applied to identify informative genomic features. The processed dataset will then be divided into training (70%) and test ing (30%) subsets. Four supervised machine learning algorithms, namely Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), will be implemented and compared to classify disease samples based on their gene expression profiles. Keywords— Genomic Data Science, Gene Expression Analysis, Cancer, Liver Disease, Machine Learning, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, Biomarker Discovery

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Nomenclature

Roman Symbols

F	complex function
F	complex function

Greek Symbols

$\imath$	unit imaginary number $\sqrt{-1}$
$\imath$	unit imaginary number $\sqrt{-1}$
π	$\simeq 3.14\ldots$
π	$\simeq 3.14\ldots$

Superscripts

j	superscript index
j	superscript index

Subscripts

0	subscript index
0	subscript index

Other Symbols

$\oint_\gamma$	integration around a curve γ
$\oint_\gamma$	integration around a curve γ

Acronyms / Abbreviations

CIF Cauchy's Integral Formula

CIF Cauchy's Integral Formula

Chapter 1

Introduction

1.1 Background

Advances in next-generation sequencing (NGS) technologies and bioinformatics have significantly transformed biomedical research by enabling the analysis of large-scale genomic data. Genomic Data Science combines computational methods, statistics, and molecular biology to extract meaningful biological information from genomic datasets. One of the most important applications of genomic data science is the identification of differentially expressed genes (DEGs) and disease-associated genetic variations, which provide valuable insights into disease mechanisms, diagnosis, prognosis, and therapeutic development. Cancer and liver disease are among the leading causes of morbidity and mortality worldwide. Although these diseases affect different organs, both are associated with complex genetic alterations and abnormal gene expression. Understanding these genomic changes is essential for identifying biomarkers and discovering novel therapeutic targets. Publicly available databases such as the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) provide researchers with high-quality genomic datasets that support comparative analyses across different diseases. This research proposes a comparative genomic data science analysis using the cancer dataset (GSE329336) and the liver disease dataset (GSE280687). By applying bioinformatics techniques, the study aims to identify differentially expressed genes, investigate disease-associated genetic variations, compare shared and disease-specific molecular signatures, and perform functional enrichment analysis. The findings are expected to contribute to a better understanding of the molecular mechanisms underlying cancer and liver disease while supporting future research in precision medicine.

1.2 Motivation

Cancer and liver disease are major health concerns that involve complex changes in gene expression. Although publicly available genomic datasets provide valuable opportunities for disease research, many existing studies focus on individual diseases rather than comparing their molecular characteristics. This limits the understanding of shared and disease-specific gene expression patterns. Therefore, this research aims to conduct a comparative genomic analysis using publicly available cancer (GSE329336) and liver disease (GSE280687) datasets. By applying bioinformatics techniques and machine learning algorithms, the study seeks to identify important gene expression patterns, compare disease characteristics, and explore potential biomarkers that may support future disease classification and precision medicine research.

1.3 Problem Statement

Despite the availability of large-scale genomic datasets in public repositories, comparative analyses between different diseases remain limited. Most existing studies focus on a single disease, making it difficult to identify common and unique genetic characteristics across multiple diseases. Furthermore, variations in gene expression and molecular pathways between cancer and liver disease have not been comprehensively investigated using publicly available datasets. The lack of integrated comparative genomic analysis limits the discovery of shared biomarkers, disease-specific genes, and biological pathways that could improve disease diagnosis and treatment. Therefore, there is a need for a comprehensive bioinformatics-based comparative study that analyzes genomic datasets from both cancer and liver disease to identify differential gene expression patterns and disease-associated genetic variations.

As shown in the figure:

- **Input:** Genomic data of cancer and liver disease
- **Processing:** Data preprocessing, feature selection, and train/test split
- **Machine Learning:** Classification algorithms, model training, and evaluation
- **Output:** Cancer and liver disease classification

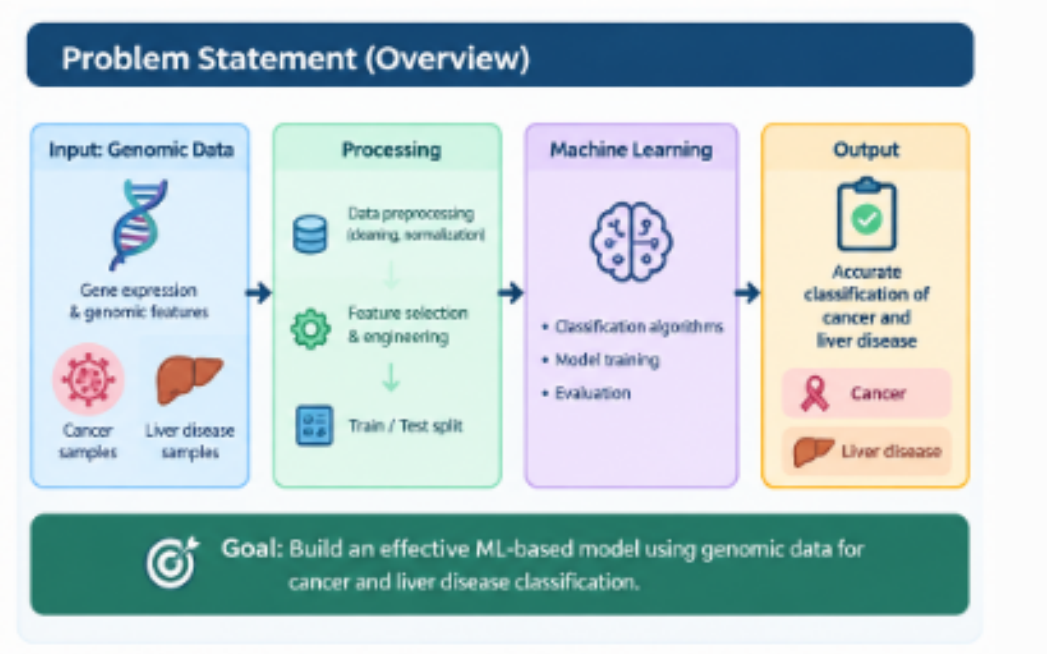


Fig. 1.1 Problem Statement

1.4 Research Challenges

To perform a comprehensive comparative genomic data science analysis of cancer and liver disease datasets for the identification of differential gene expression patterns and disease-associated genetic variations.

- **Data Collection:** To collect and preprocess publicly available genomic datasets from the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO).
- **Gene Identification:** To identify differentially expressed genes (DEGs) in the cancer and liver disease datasets.
- **Comparative Analysis:** To compare shared and disease-specific gene expression patterns between the two diseases.
- **Variation Investigation:** To investigate disease-associated genetic variations using bioinformatics approaches.
- **Pathway Enrichment:** To perform Gene Ontology (GO) and pathway enrichment analyses to identify significant biological processes and molecular pathways.

- **Biomarker Discovery:** To identify potential biomarkers and therapeutic targets that may contribute to improved diagnosis and treatment of cancer and liver disease.

Chapter 2

Related Work

The rapid development of genomic data science has transformed biomedical research by enabling the analysis of large-scale genomic datasets to understand disease mechanisms at the molecular level. Advances in next-generation sequencing (NGS) and RNA sequencing (RNA-Seq) technologies have made it possible to identify differentially expressed genes (DEGs), detect disease-associated genetic variations, and discover potential biomarkers for diagnosis and treatment. Numerous computational methods and bioinformatics tools have been developed to analyze these datasets efficiently. Rosati et al. [13] reviewed several differential gene expression analysis pipelines and emphasized that selecting appropriate preprocessing, normalization, and statistical analysis methods is essential for obtaining reliable and reproducible results. Similarly, Costa-Silva et al. [4] provided a comprehensive overview of computational approaches for RNA-Seq data analysis and discussed the strengths and limitations of commonly used DEG identification methods. The availability of publicly accessible genomic repositories has further accelerated ge nomic research. The National Center for Biotechnology Information (NCBI) maintains the Gene Expression Omnibus (GEO), one of the world's largest repositories of gene expres sion datasets, allowing researchers to perform large-scale computational analyses without conducting laboratory experiments [8]. In this proposed research, two publicly available GEOdatasets, GSE329336 [10] and GSE280687 [9], will be analyzed to investigate gene expression differences between cancer and liver disease. Cancer genomics has received significant attention because of the complexity and hetero geneity of cancer cells. Fan et al. [5] demonstrated that single-cell transcriptomics enables researchers to investigate gene expression at the cellular level, revealing tumor heterogeneity that cannot be observed using conventional bulk RNA sequencing. Likewise, Li et al. [6] conducted a systematic review on the applications of single-cell RNA sequencing in cancer research and concluded that this technology has become an essential tool for understanding tumor progression, immune cell

interactions, and personalized therapeutic strategies. Fur thermore, Ma et al. [7] highlighted that integrating multiple omics technologies, including genomics, transcriptomics, proteomics, and metabolomics, provides a more comprehen sive understanding of cancer biology and facilitates the discovery of clinically relevant biomarkers. Genomic studies have also significantly contributed to understanding liver diseases. Oh et al. [11] discussed how genome sequencing technologies have enabled researchers to identify novel biomarkers associated with chronic liver diseases and hepatocellular carcinoma. Their study emphasized that genomic analysis can improve early diagnosis and support the development of targeted therapies. Similarly, Ramachandran et al. [12] applied single-cell genomics and spatial transcriptomics to investigate liver physiology and disease progression, demonstrating that cellular diversity plays a critical role in liver pathology. Earlier studies by Shibata and Aburatani [14] also revealed that comprehensive genomic profiling has greatly improved the understanding of genetic mutations involved in liver cancer development. The continuous advancement of bioinformatics has introduced new analytical methods for processing increasingly complex genomic datasets. Das Adhikari et al. [1] reviewed recent developments in spatial transcriptomics and explained how spatially variable gene detection provides additional biological context beyond traditional gene expression analysis. Yu et al. [15] addressed another major challenge in genomic research by investigating batch effects in large-scale omics datasets. Their study demonstrated that effective batch correction techniques significantly improve the reliability and reproducibility of downstream analyses. RNA sequencing has also become an important tool in clinical genomics. Byron et al. [3] discussed the opportunities and challenges of translating RNA sequencing into clinical diagnostics and highlighted its potential for personalized medicine. Similarly, Anders et al. [2] introduced count-based differential expression analysis methods using R and Bioconductor, which have become widely adopted for identifying differentially expressed genes in RNA-Seq datasets and remain standard approaches in genomic research. Although previous studies have successfully investigated genomic characteristics of cancer and liver disease independently, relatively few studies have performed a comprehensive comparative analysis using publicly available datasets from both diseases. Most research focuses on identifying biomarkers or molecular pathways within a single disease rather than comparing disease-specific and shared gene expression patterns across multiple conditions. Therefore, there remains a research gap in comparative genomic data science that integrates standardized bioinformatics workflows with publicly available datasets. The proposed research aims to address this gap by performing a comprehensive com parative genomic analysis of the cancer dataset GSE329336 and the liver disease dataset GSE280687. Throughdifferential gene expression analysis, investigation of disease-associated genetic

variations, and functional enrichment analysis, the study seeks to identify common and disease-specific molecular signatures that may contribute to biomarker discovery, preci sion medicine, and future therapeutic development.

Chapter 3

Proposed Methodology

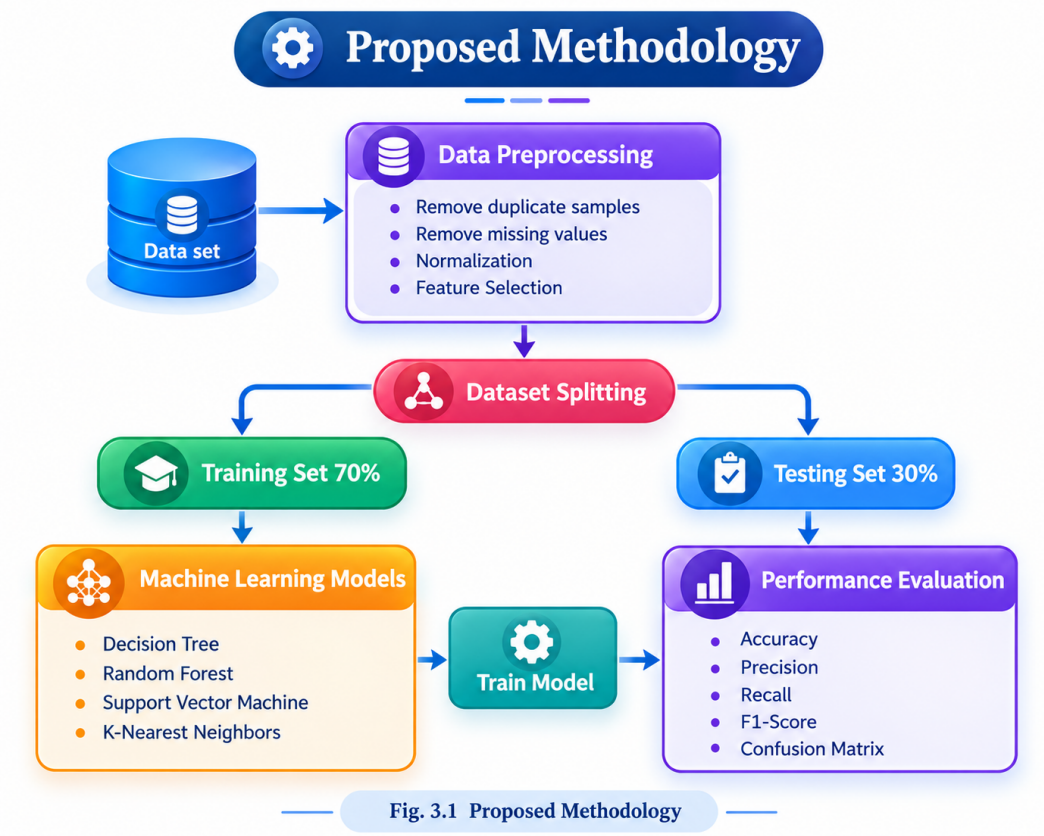


Fig. 3.1 Proposed Methodology

This chapter presents the proposed methodology for conducting a comparative genomic data science analysis of cancer and liver disease datasets. The methodology consists of several sequential stages, including dataset collection, data preprocessing, dataset splitting, machine learning model development, model training, and performance evaluation[cite: 6]. The primary objective of this methodology is to identify disease-associated gene expression

patterns and evaluate the effectiveness of different machine learning algorithms for disease classification[cite: 6]. Figure 3.1 illustrates the overall workflow of the proposed research[cite: 6].

3.1 Dataset

This research utilizes publicly available gene expression datasets obtained from the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO)[cite: 6]. Two datasets are selected for comparative genomic analysis[cite: 6]:

- **Cancer Dataset:** GSE329336[cite: 6]
- **Liver Disease Dataset:** GSE280687[cite: 6]

These datasets contain gene expression profiles collected from patients diagnosed with cancer and liver disease[cite: 6]. Since the datasets are publicly available and experimentally validated, they provide reliable genomic information for comparative analysis without requiring additional laboratory experiments[cite: 6].

3.2 Data Preprocessing

Data preprocessing is an essential step that improves data quality before applying machine learning algorithms[cite: 6]. Raw genomic datasets often contain redundant records, missing values, and inconsistencies that may negatively affect model performance[cite: 6]. Therefore, several preprocessing techniques will be performed[cite: 6].

3.2.1 Duplicate Samples

Duplicate samples can introduce bias into the learning process and affect classification accuracy[cite: 6]. Therefore, duplicate observations will be identified and removed to ensure that each biological sample appears only once in the dataset[cite: 6].

3.2.2 Handle Missing Values

Missing values are common in genomic datasets due to sequencing limitations or incomplete measurements[cite: 6]. Missing records will be identified and appropriately handled to improve data consistency and reduce potential analytical errors[cite: 6].

3.2.3 Data Normalization

Gene expression values often have different numerical scales[cite: 7]. Normalization will be applied to transform the data into a comparable range, ensuring that genes contribute equally during model training and preventing algorithms from being dominated by large-valued features[cite: 7].

3.2.4 Feature Selection

Gene expression datasets usually contain thousands of genes, many of which are irrelevant to disease classification[cite: 7]. Feature selection techniques will be employed to identify the most informative genes while removing redundant and low-impact features[cite: 7]. This process reduces computational complexity, minimizes overfitting, and improves model performance[cite: 7].

3.2.5 Dataset Splitting

After preprocessing, the cleaned dataset will be divided into two subsets for machine learning experiments[cite: 7]. The training dataset will contain 70% of the samples and will be used to train the machine learning models[cite: 7]. The remaining 30% of the data will be used as the testing dataset to evaluate the predictive performance of the trained models using unseen samples[cite: 7]. This data partitioning strategy helps ensure a fair and unbiased evaluation of each classifier[cite: 7].

3.3 Machine Learning Models

Four supervised machine learning algorithms will be implemented and compared in this research[cite: 7].

3.3.1 Decision Tree

The Decision Tree algorithm constructs a hierarchical tree structure to classify gene expression samples based on decision rules[cite: 7]. It is simple to interpret and provides clear insights into the decision-making process[cite: 7].

3.3.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting[cite: 7]. It is widely used in genomic data analysis because of its robustness when dealing with high-dimensional biological data[cite: 7].

3.3.3 Support Vector Machine (SVM)

Support Vector Machine is one of the most effective classification algorithms for gene expression analysis[cite: 8]. It identifies an optimal decision boundary that maximizes the separation between different disease classes and performs well with high-dimensional datasets[cite: 8].

3.3.4 K-Nearest Neighbors (KNN)

K-Nearest Neighbors classifies samples according to the labels of their nearest neighboring samples based on feature similarity[cite: 8]. It is a simple yet effective algorithm that serves as a useful baseline for performance comparison[cite: 8].

3.4 Model Training

The preprocessed training dataset will be used to train the four machine learning models[cite: 8]. During this phase, each algorithm will learn the relationship between gene expression patterns and disease categories[cite: 8]. After training, the developed models will be applied to the testing dataset to predict disease classifications[cite: 8]. The performance of each algorithm will then be compared to determine the most effective model for genomic disease classification[cite: 8].

3.5 Performance Evaluation

The effectiveness of the proposed machine learning models will be assessed using standard classification performance metrics[cite: 8].

- **Accuracy:** Measures the overall percentage of correctly classified samples[cite: 8].
- **Precision:** Evaluates the proportion of correctly predicted positive samples among all predicted positive samples[cite: 8].

- **Recall:** Measures the ability of the model to correctly identify actual positive samples[cite: 8].
- **F1-Score:** Provides the harmonic mean of precision and recall, offering a balanced measure of classification performance[cite: 8].
- **Confusion Matrix:** Presents the number of correctly and incorrectly classified samples for each disease category, providing a detailed evaluation of model performance[cite: 8].

These evaluation metrics will be calculated for each machine learning model, and the results will be compared to identify the best-performing classifier for comparative genomic disease analysis[cite: 9].

3.6 Summary

The proposed methodology provides a systematic framework for conducting comparative genomic data analysis using publicly available cancer and liver disease datasets[cite: 9]. Beginning with data collection and preprocessing, the methodology proceeds through dataset splitting, machine learning model development, model training, and performance evaluation[cite: 9]. The comparative analysis of Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors will enable the identification of the most effective classification model for genomic disease prediction[cite: 9]. The findings are expected to support biomarker discovery and contribute to future research in genomic data science and precision medicine[cite: 9].

Chapter 4

Literature Review and Research Gap Analysis

Table 4.1 presents a summary of previous studies related to genomic data analysis, liver disease biomarkers, differential gene expression (DEG) analysis, and genomic data reliability. The reviewed studies highlight important contributions to understanding disease-associated genes and molecular characteristics. However, several research gaps remain in comparative genomic analysis and disease classification.

Oh et al. (2020) investigated potential biomarkers associated with liver disease and highlighted the importance of genomic analysis in identifying disease-related biomarkers. However, the comparison of genomic characteristics between cancer and other liver diseases was not the primary focus of their study.

Ramachandran et al. (2020) explored liver physiology and disease biology using single-cell genomics and spatial transcriptomics. Their research provided insights into liver cellular diversity and disease progression. Nevertheless, comparative classification of cancer and liver disease was not the central objective of their work.

Rosati et al. (2024) reviewed bioinformatics pipelines and computational tools for differential gene expression analysis and biomarker identification. Their work emphasized the importance of appropriate analytical methods for reliable genomic analysis. However, further investigation is needed to apply these approaches in integrated comparative studies across different diseases.

Shibata and Aburatani (2014) examined the genomic characteristics of liver cancer and contributed to a better understanding of genetic mutations involved in cancer development. However, their research primarily focused on liver cancer genomics rather than comparative classification across broader cancer and liver disease categories.

Yu et al. (2024) investigated batch effects in large-scale omics studies and discussed methods for improving data reliability and reproducibility. Although batch effect correction is important for genomic data analysis, comparative genomic classification of cancer and liver disease was not the main focus of their study.

Overall, the reviewed literature demonstrates significant progress in genomic biomarker discovery, liver disease research, differential gene expression analysis, and data quality improvement. However, the comparison of shared and disease-specific gene expression patterns across cancer and liver disease remains an important research direction. Therefore, the proposed research aims to analyze publicly available genomic datasets and evaluate machine learning algorithms for disease classification. This study seeks to contribute to comparative genomic data science and the identification of potential disease-associated biomarkers.

Author & Year	**Research Focus**	**Key Findings**	**Research Gap**
Oh et al. (2020)[cite: 4]	Liver Biomarkers[cite: 4]	Identified genomic liver biomarkers[cite: 4]	Cancer comparison missing[cite: 4]
Ramachandran et al. (2020)[cite: 4]	Liver Genomics[cite: 4]	Explained liver cellular diversity[cite: 4]	Comparative classification absent[cite: 4]
Rosati et al. (2024)[cite: 4]	DEG Pipelines[cite: 4]	Reviewed biomarker identification tools[cite: 4]	Cross-disease analysis limited[cite: 4]
Shibata & Aburatani (2014)[cite: 4]	Liver Cancer Genomics[cite: 4]	Improved understanding of mutations[cite: 4]	Single disease focus[cite: 4]
Yu et al. (2024)[cite: 4]	Batch Effect Correction[cite: 4]	Increased omics data reliability[cite: 4]	Comparative genomic classification absent[cite: 4]

Table 4.1 Summary of Literature Review

Chapter 5

Genomic Data Science Frameworks: Understanding Big Data Challenges in Gene Expression Analysis

5.1 Introduction

Gene expression analysis for cancer and liver disease classification sits at the intersection of genomics and data science. As sequencing technologies advance, the volume, velocity, and variety of genomic data have grown exponentially. Understanding these challenges through established data science frameworks is essential for designing robust machine learning pipelines.

This chapter introduces the key frameworks used in genomic data science — the **3V / 5V framework** and the **4M process framework** — and discusses how they apply to gene expression-based disease classification.

5.2 The Biomedical Data Science Umbrella

Genomics is one of the most data-intensive subfields within the broader biomedical data science landscape. Figure 7.1 illustrates how different omics disciplines (genomics, transcriptomics, proteomics, metabolomics, etc.) converge with imaging, wearable devices, and health records to form a unified data science ecosystem.

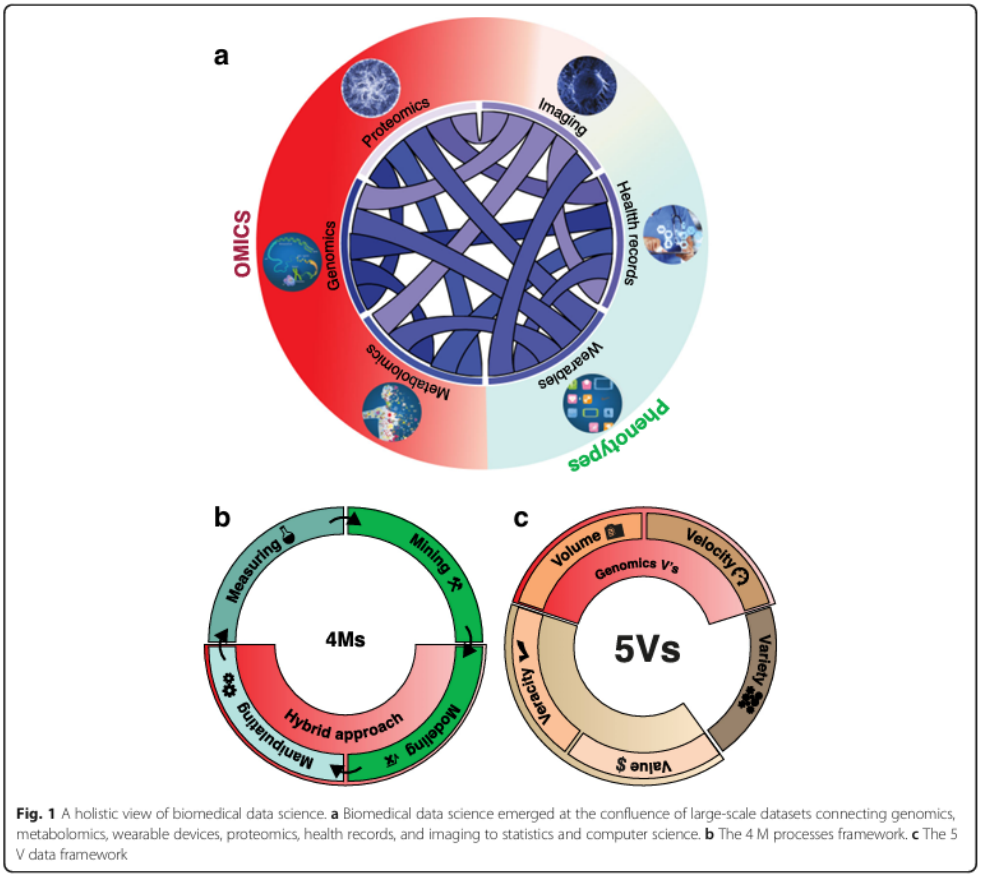


Fig. 1 A holistic view of biomedical data science. **a** Biomedical data science emerged at the confluence of large-scale datasets connecting genomics, metabolomics, wearable devices, proteomics, health records, and imaging to statistics and computer science. **b** The 4 M processes framework. **c** The 5 V data framework

Fig. 5.1 Holistic view of Biomedical Data Science

5.2.1 The 3V and 5V Frameworks in Genomics

The classic **3V framework** (Volume, Velocity, Variety) helps characterize the challenges of big genomic data. Later extensions added **Value** and **Veracity**, forming the **5V framework**.

Volume: Genomic data volume is growing faster than most other scientific domains. While currently smaller than Earth science datasets, projections show genomics may surpass social media and astronomy data in the coming decade if current trends continue.

Velocity: Human genome sequencing that once took 13 years can now be completed in less than 24 hours. This rapid data generation requires equally fast preprocessing and analysis pipelines.

Variety: Beyond raw nucleotide sequences, gene expression studies involve diverse data types: microarray intensities, RNA-seq counts, clinical phenotypes, and imaging data. This heterogeneity makes feature engineering and model generalization particularly challenging.

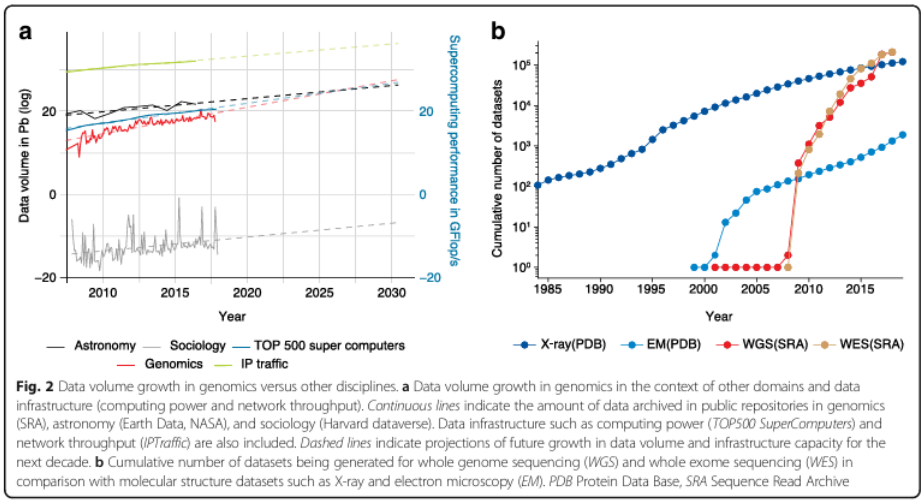


Fig. 2 Data volume growth in genomics versus other disciplines. **a** Data volume growth in genomics in the context of other domains and data infrastructure (computing power and network throughput). *Continuous lines* indicate the amount of data archived in public repositories in genomics (SRA), astronomy (Earth Data, NASA), and sociology (Harvard dataverse). Data infrastructure such as computing power (*TOP500 SuperComputers*) and network throughput (*IPTraffic*) are also included. *Dashed lines* indicate projections of future growth in data volume and infrastructure capacity for the next decade. **b** Cumulative number of datasets being generated for whole genome sequencing (*WGS*) and whole exome sequencing (*WES*) in comparison with molecular structure datasets such as X-ray and electron microscopy (*EM*). *PDB* Protein Data Base, *SRA* Sequence Read Archive

Fig. 5.2 Data Volume Growth Comparison (Genomics, Astronomy, Sociology, Supercomputing power, and Network traffic from 2010–2030)

5.3 The 4M Process Framework

Beyond characterizing data, the **4M framework** (Measurement → Mining → Modeling → Manipulation) describes the complete scientific process in systems biology and genomics:

1. **Measurement** – Generating gene expression profiles.
2. **Mining** – Feature selection and pattern discovery using machine learning.
3. **Modeling** – Building predictive classifiers (Decision Tree, Random Forest, SVM, KNN).
4. **Manipulation** – Validating models and generating new biological hypotheses.

This hybrid approach of combining data-driven mining with biological modeling is particularly powerful for disease classification tasks.

5.4 Implications for Cancer and Liver Disease Classification

High dimensionality (tens of thousands of genes vs. limited samples), batch effects, missing values, and class imbalance are direct consequences of the V and M frameworks. The preprocessing and feature selection steps proposed in this thesis directly address these challenges.

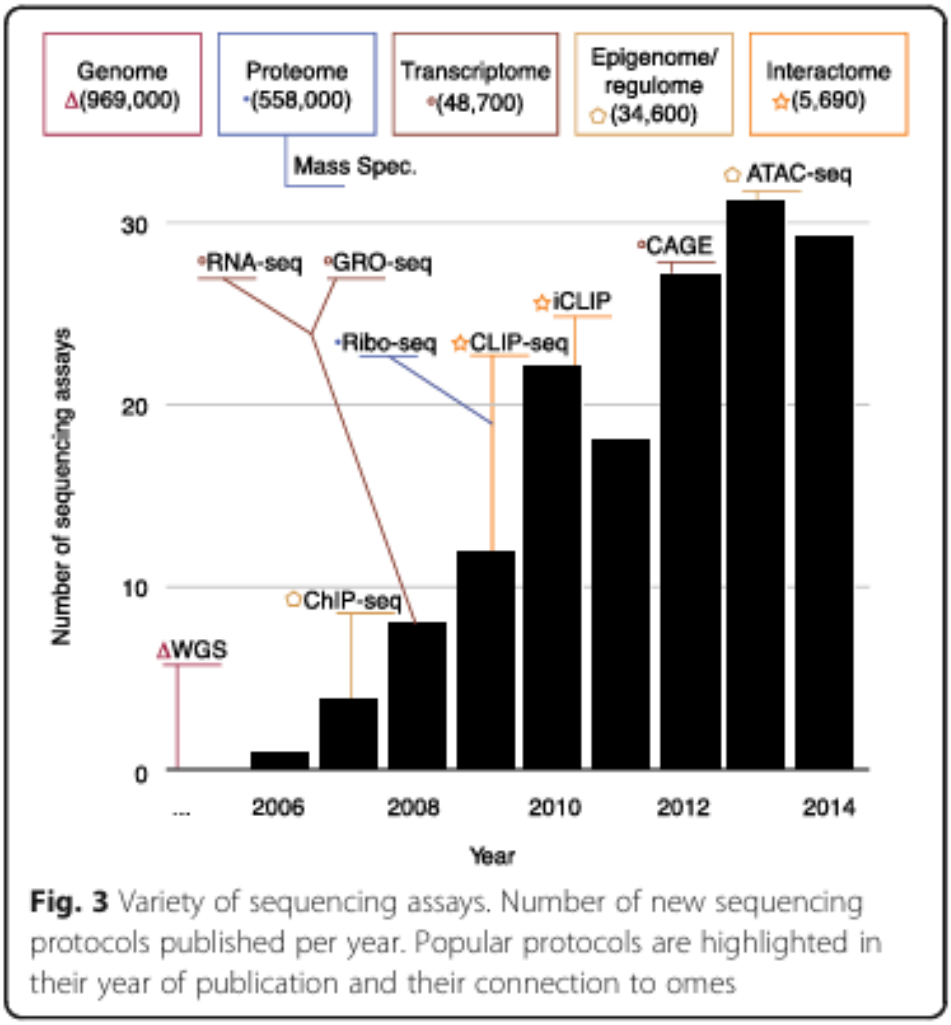


Fig. 3 Variety of sequencing assays. Number of new sequencing protocols published per year. Popular protocols are highlighted in their year of publication and their connection to omes

Fig. 5.3 The 5V Framework Applied to Genomics highlighting Volume, Velocity, Variety, Value, and Veracity

5.5 Summary

Understanding genomic data through the 3V/5V and 4M frameworks provides a solid theoretical foundation for designing effective machine learning pipelines for disease classification.

Chapter 6

Privacy, Security, and Ethical Considerations in Genomic Machine Learning

6.1 Introduction

Gene expression datasets, even when de-identified, carry significant privacy risks because genomic information is unique, persistent, and can reveal information about family members[cite: 129]. As machine learning models are trained on such sensitive data (often using cloud resources), privacy and security become critical concerns[cite: 128].
This chapter discusses the major privacy threats, secure computation techniques, and ethical considerations relevant to gene expression-based disease classification.

6.2 Privacy Risks in Genomic Data

Even after removing explicit identifiers, genomic data can be re-identified through various advanced techniques[cite: 129]:

- Side-channel attacks[cite: 129]
- Completion attacks[cite: 129]
- Genealogical triangulation[cite: 129]

Once released, genomic data remains permanently public and can affect not only the individual but also their relatives[cite: 129].

6.3 Secure Computation Techniques

Two major approaches have emerged for protecting genomic data analytics[cite: 130]:

1. **Homomorphic Encryption (HME):** Allows computation directly on encrypted data[cite: 130]. Three main types exist:
   - **Partial HME:** Specialized in a single operation (addition or multiplication), making it efficient but limited[cite: 130].
   - **Fully Homomorphic Encryption (FHE):** Supports both operations to implement any function theoretically, but is computationally and storage intensive[cite: 130].
   - **Levelled HME:** A practical compromise that allows both operations for a pre-specified, limited number of iterations with reasonable efficiency[cite: 130].
2. **Secure Multi-Party Computation (SMC):** Enables multiple institutions to jointly analyze data without sharing raw individual-level records[cite: 128, 130].

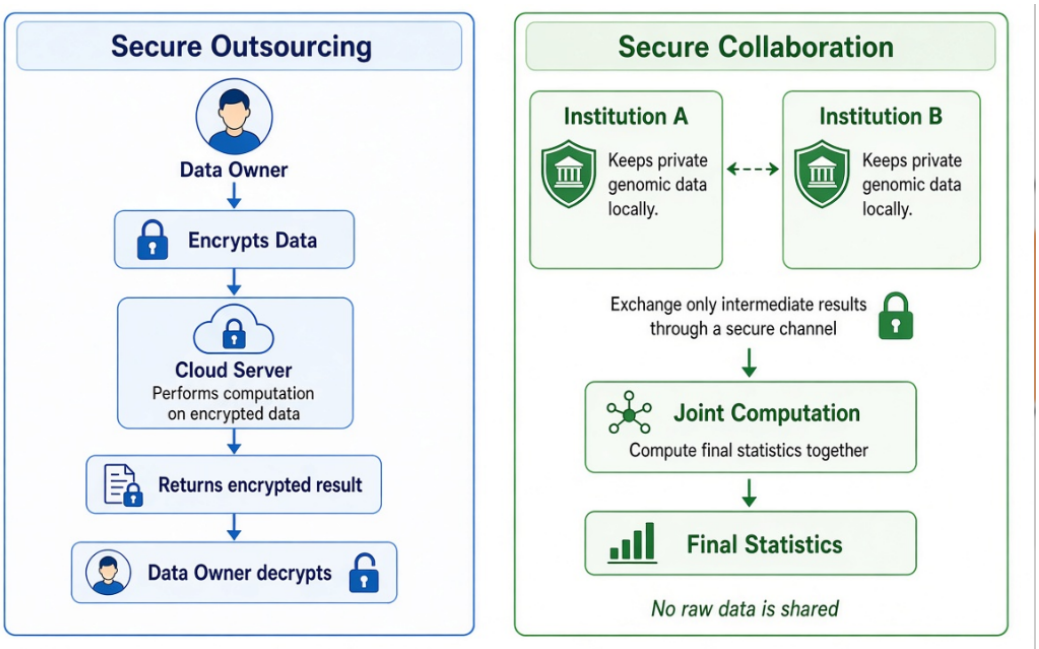


Fig. 6.1 Secure Outsourcing vs Secure Collaboration

6.4 Practical Challenges and Current Limitations

While secure techniques can already handle certain tasks (e.g., Hamming distance calculation on 100,000 bases in minutes)[cite: 131], more complex analyses such as

Genome-Wide Association Studies (GWAS) still face significant computational and communication overhead[cite: 131, 133].

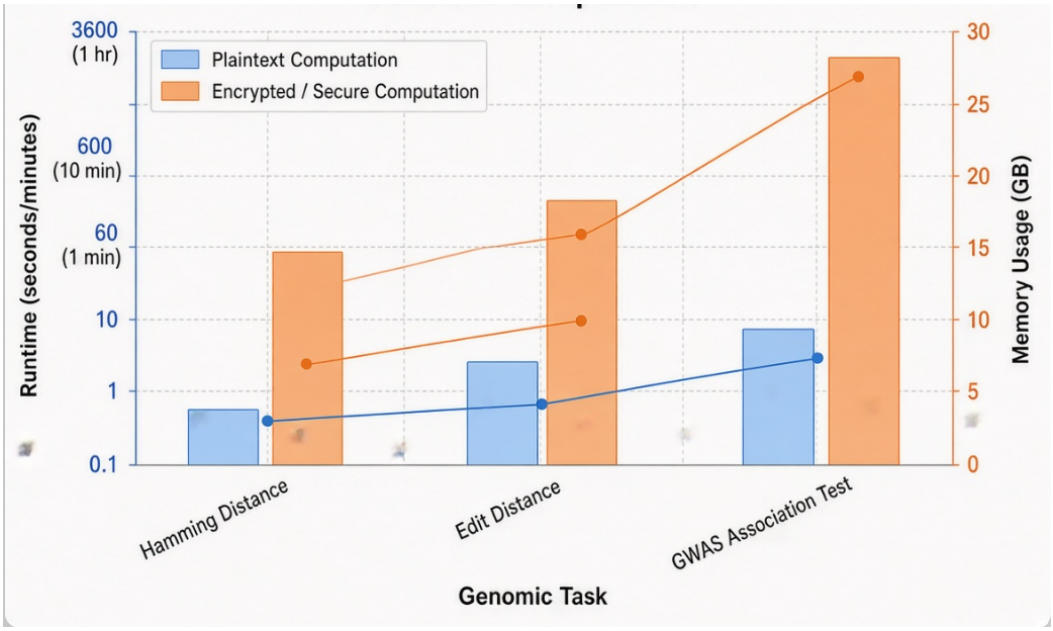


Fig. 6.2 Performance vs Privacy Trade-off in Genomic Computation

6.5 Ethical and Policy Considerations

Key ethical and regulatory challenges in genomic data processing include:

- Data ownership and consent[cite: 135]
- Secondary use of data beyond original research scope[cite: 128]
- Institutional policies versus commercial cloud environments[cite: 128, 129]
- Long-term privacy risks as re-identification techniques improve[cite: 129]

Researchers remain responsible for data security even when using public cloud services, as outlined by policies such as the NIH Genome Data Sharing (GDS) policy[cite: 129].

6.5.1 Relevance to the Present Study

Although this thesis uses publicly available GEO datasets, the privacy principles discussed here are essential for any future clinical translation of the developed models. Secure

computation techniques should be considered when deploying gene expression classifiers in real healthcare settings.

6.6 Summary

Protecting genomic data while enabling powerful machine learning analysis remains an active research challenge. Balancing utility and privacy is crucial for the responsible advancement of precision medicine.

Chapter 7

Experimental Work

This chapter describes the experimental procedures that will be followed to evaluate the effectiveness of the proposed machine learning models for comparative genomic data analysis. The experiments will be conducted using publicly available gene expression datasets of cancer and liver disease obtained from the NCBI Gene Expression Omnibus (GEO). The primary objective is to investigate the classification performance of different machine learning algorithms and identify the most suitable model for disease prediction based on genomic data.

7.1 Experimental Environment

The experiments will be conducted using a computer with sufficient computational resources to process high-dimensional genomic datasets. Python will be used as the primary programming language because of its extensive support for bioinformatics and machine learning applications. The implementation will be carried out using Jupyter Notebook, while several scientific computing and machine learning libraries will be utilized throughout the experiments. The major software tools and libraries include:

- Python
- NumPy
- Pandas
- Scikit-learn
- Matplotlib

- Seaborn
- Colab

These tools provide efficient support for data preprocessing, machine learning model development, visualization, and performance evaluation.

7.2 Experimental Procedure

Initially, the cancer dataset (GSE329336) and the liver disease dataset (GSE280687) will be downloaded from the NCBI GEO database. The datasets will undergo preprocessing to improve data quality before model development. Duplicate samples will be removed, missing values will be handled, the data will be normalized, and feature selection techniques will be applied to retain only the most informative genomic features. Following preprocessing, the dataset will be divided into two subsets. Approximately 70% of the data will be used for model training, while the remaining 30% will be reserved for testing. This partitioning strategy enables an unbiased evaluation of model performance on unseen data. Four supervised machine learning algorithms will then be implemented, namely Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Each model will be trained using the same training dataset to ensure a fair comparison. After training, the models will classify the testing samples based on their gene expression profiles.

7.3 Performance Evaluation

The performance of each machine learning model will be evaluated using several widely accepted classification metrics:

- **Accuracy:** Measures the overall percentage of correctly classified samples.
- **Precision:** Evaluates the proportion of correctly predicted positive samples among all predicted positive samples.
- **Recall:** Measures the capability of the classifier to correctly identify actual positive samples.
- **F1-Score:** Provides a balanced evaluation by combining precision and recall into a single metric.

- **Confusion Matrix:** Presents the detailed classification results by showing the numbers of true positives, true negatives, false positives, and false negatives.

The evaluation metrics obtained from each classifier will be compared to determine the model that provides the highest classification performance for genomic disease prediction.

7.4 Comparative Analysis

A comparative analysis will be conducted among the four machine learning algorithms. Their classification performance, computational efficiency, and prediction capability will be analyzed using the evaluation metrics. The comparison will identify the strengths and limitations of each algorithm when applied to high-dimensional genomic datasets. The experimental findings are expected to determine which machine learning model performs best for distinguishing cancer and liver disease samples based on gene expression patterns. These results may also provide valuable insights for future biomarker discovery and precision medicine research.

7.4.1 Summary

This chapter presents the experimental design proposed for evaluating the performance of the selected machine learning algorithms. The experiments include dataset preprocessing, model training, classification, and performance evaluation. The comparative analysis of Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors is expected to identify the most effective model for genomic disease classification and support the objectives of this research.

Chapter 8

Conclusion and Future Direction

8.1 Conclusion

This research proposal presents a comprehensive framework for the comparative genomic data science analysis of cancer and liver disease using publicly available gene expression datasets from the NCBI Gene Expression Omnibus (GEO). The proposed methodology integrates data preprocessing, machine learning techniques, and performance evaluation to identify disease-associated gene expression patterns and improve disease classification. The study will utilize two genomic datasets, GSE329336 for cancer and GSE280687 for liver disease. After preprocessing the datasets by removing duplicate samples, handling missing values, normalizing the data, and selecting informative features, four machine learning algorithms—Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—will be trained and evaluated. Their performance will be assessed using standard evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The comparative analysis of these machine learning models is expected to determine the most effective classifier for genomic disease prediction. Furthermore, the findings may contribute to the identification of important gene expression patterns that can support biomarker discovery and provide valuable insights into the molecular mechanisms of cancer and liver disease. Overall, this research aims to contribute to the growing field of genomic data science by applying computational approaches to improve disease analysis and support future precision medicine research

8.2 Future Direction

Several opportunities exist to extend this research in the future. First, additional genomic datasets from different diseases can be incorporated to perform broader comparative analyses and improve the generalizability of the proposed framework. Second, advanced feature selection and dimensionality reduction techniques may be explored to enhance classification performance and reduce computational complexity. Future studies may also investigate deep learning approaches, such as Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Graph Neural Networks (GNNs), for analyzing complex genomic data. In addition, integrating multi-omics data, including transcriptomics, proteomics, metabolomics, and epigenomics, could provide a more comprehensive understanding of disease mechanisms. Finally, the identified biomarkers and disease-associated gene expression patterns may be validated using independent clinical datasets or experimental laboratory studies. Such validation would increase the reliability of the findings and support the development of personalized diagnostic tools and targeted therapeutic strategies for cancer and liver disease.

6. Working Schedule

MONTH	ACTIVITY	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER
July (July 1 – July 31)	• Finalize research problem and objectives • Conduct extensive literature review • Select datasets or plan data collection strategy						
August (Aug 1 – Aug 31)	• Dataset acquisition and annotation • Begin preprocessing pipeline • Start building baseline models (e.g., BERT, mBERT)						
September (Sep 1 – Sep 30)	• Implement multilingual transformers (XLM-RoBERTa etc.) • Fine-tune models • Integrate multi-label classification logic						
October (Oct 1 – Oct 31)	• Perform model evaluation with Micro/Macro-F1, Hamming Loss • Run comparative experiments • Conduct error analysis						
November (Nov 1 – Nov 25)	• Improve models using ensemble or label correlation-aware methods • Finalize results and figures • Start writing thesis/report						
End of November (Nov 26 – Dec 1)	• Complete final report writing • Format bibliography and appendices • Submit project report / prepare for presentation						

Fig. 8.1 Proposed Research Working Schedule and Gantt Chart

References

[1] Adhikari, S. D. et al. (2024). Recent advances in spatially variable gene detection in spatial transcriptomics. *Computational and Structural Biotechnology Journal*.

[2] Anders, S., McCarthy, D. J., Chen, Y., Okoniewski, M., Smyth, G. K., Huber, W., and Robinson, M. D. (2013). Count-based differential expression analysis of rna sequencing data using r and bioconductor. *Nature Protocols*, 8(9):1765–1786.

[3] Byron, S. A., Keuren-Jensen, K. R. V., Engelthaler, D. M., Carpten, J. D., and Craig, D. W. (2016). Translating rna sequencing into clinical diagnostics: Opportunities and challenges. *Nature Reviews Genetics*, 17(5):257–271.

[4] Costa-Silva, J., Domingues, D. S., Hungria, M., and Lopes, F. M. (2021). Computational methods for differentially expressed gene analysis from rna-seq: An overview. *arXiv preprint arXiv:2109.03625*.

[5] Fan, J., Slowikowski, K., and Zhang, F. (2020). Single-cell transcriptomics in cancer: Computational challenges and opportunities. *Experimental and Molecular Medicine*, 52:1452–1465.

[6] Li, L. et al. (2021). Applications of single-cell rna sequencing in cancer research: A systematic review. *Journal of Experimental and Clinical Cancer Research*, 40.

[7] Ma, W. et al. (2024). A review on trends in development and translation of omics signatures in cancer. *Computational and Structural Biotechnology Journal*.

[8] National Center for Biotechnology Information (2025a). Gene expression omnibus (geo).

[9] National Center for Biotechnology Information (2025b). Gse280687: Liver disease gene expression dataset.

[10] National Center for Biotechnology Information (2025c). Gse329336: Cancer gene expression dataset.

[11] Oh, S., Jo, Y., Jung, S., Yoon, S., and Yoo, K. H. (2020). From genome sequencing to the discovery of potential biomarkers in liver disease. *BMB Reports*, 53(6):299–310.

[12] Ramachandran, P. et al. (2020). Single-cell genomics and spatial transcriptomics in liver physiology and disease biology. *Journal of Hepatology*, 73(5):1219–1230.

[13] Rosati, D., Palmieri, M., Brunelli, G., Morrione, A., Iannelli, F., Frullanti, E., and Giordano, A. (2024). Differential gene expression analysis pipelines and bioinformatic tools for the identification of specific biomarkers: A review. *Computational and Structural Biotechnology Journal*, 23:1154–1168.

[14] Shibata, T. and Aburatani, H. (2014). Exploration of liver cancer genomes. *Nature Reviews Gastroenterology and Hepatology*, 11(6):340–349.

[15] Wang, H., Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., Chandak, P., Liu, S., Van Katwyk, P., Deac, A., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620(7972):47–60.

[16] Yu, Y. et al. (2024). Assessing and mitigating batch effects in large-scale omics studies. *Genome Biology*, 25.